Problem D
Mulk Ke Wasee Tar Mafad Mein

Time limit: 3 seconds

The provincial government has been suspended. The public has been assured that its mandate remains respected. It is merely being respected from a different office.

The outgoing administration must now vacate the premises. Statements have been issued, objections have been recorded, and someone has finally asked the practical question:

"Gaariyon ka intezam kis ne karna tha?"

Unfortunately, that question was addressed to you.

There are n outgoing babus waiting in a line. Babu i carries a_i units of luggage. Their order must be preserved. After all, protocol is the one arrangement that can never be suspended.

Normally, the roads would be lined with containers, and some babus might find themselves on a different journey altogether. For this problem, assume the roads are clear.

Divide the line into nonempty consecutive groups, assigning each group to one hired vehicle. Every babu must travel exactly once, with all their luggage. There are limits to austerity.

White Corollas and black Vigos are available in any quantity. A white Corolla has luggage capacity C_c and costs P_c to hire. A black Vigo has luggage capacity C_v and costs P_v. Either type may be larger or cheaper. The procurement committee would have preferred to know this before approving the bill. The briefcase under the table settled the matter.

Every group's total luggage must fit within its vehicle's capacity.

The new administration has also announced an austerity policy: each vehicle must leave with at least half its luggage capacity in use. At most one vehicle in the entire convoy may carry less than half its capacity. This exception may occur anywhere in the convoy. Exactly half-full is acceptable.

The policy applies equally to everyone. The exception has been approved separately. Do not let any babu tell you otherwise.

Find the minimum total hiring cost. If no valid arrangement exists, print -1.

You must work with the input provided. You may not suspend the constraints and appoint a more cooperative test case. Above your pay grade.

Input

The first line contains one integer n ($1 \le n \le 2000$).

The second line contains four integers C_c, P_c, C_v and P_v ($1 \le C_c, P_c, C_v, P_v \le 10^9$), in that order.

The third line contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$).

Output

Print one integer: the minimum total hiring cost, or -1 if no valid transportation is possible.

Sample Input 1	Sample Output 1
5 3 2 8 3 3 3 8 2 3	9

Sample Input 2	Sample Output 2
3 4 10 4 10 4 4 1	30

Sample Input 3	Sample Output 3
4 4 10 4 10 4 1 4 1	-1